as in the esterified form (Triglyceride) in blood. Fatty acids are the immediate source for oxidation of fats in various tissues viz. liver, adipose tissue, muscles, heart, kidney, brain, lungs and testes.

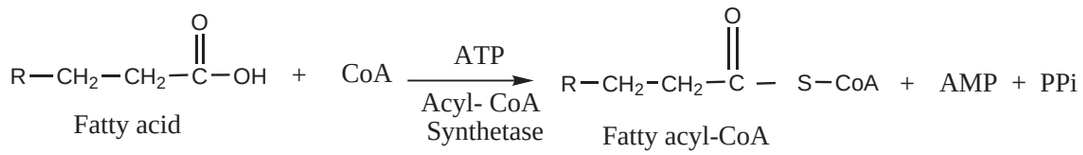
The fatty acids in the body are mostly oxidized by β-oxidation in the mitochondria. β-oxidation may be defined as the oxidation of fatty acids on the β-carbon atom. This results in the sequential removal of a two-carbon fragment, acetyl CoA as described below.

5.3.1 β-Oxidation

The β-oxidation of fatty acids involves three stages namely activation of fatty acids, transport of activated fatty acids to mitochondria followed by and the oxidation of the activated fatty acids:

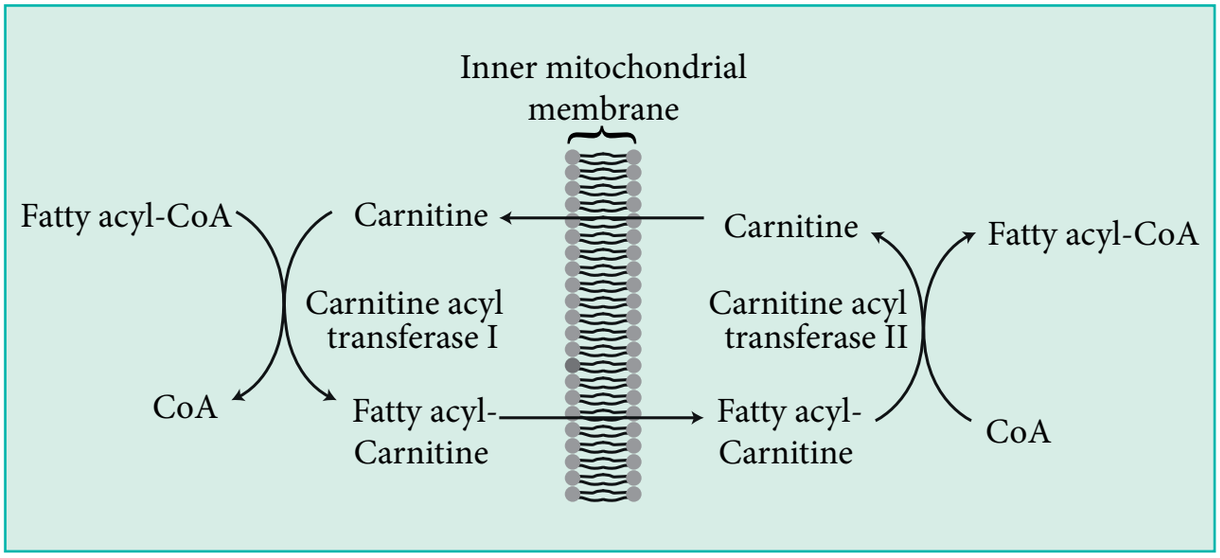
i. Activation of fatty acids occurring in the cytosol

Before the β-oxidation begins, the fatty acids are activated in a reaction with ATP and CoA to form fatty acyl CoA.



ii. Transport of fatty acids into mitochondria

Though fatty acids are activated for oxidation in cytosol, they are oxidized in the mitochondria. Since the mitochondrial membrane is impermeable for long chain fatty acids, they are transported by a special carrier molecule called carnitine.



iii. β-Oxidation in the mitochondrial matrix

Once inside the mitochondrial matrix, fatty acyl CoA is oxidized with β-carbon oxidation and the long chain acyl CoA is subjected to a repeated 4 step process which removes two carbon chains from the chain until last two carbon fragment is obtained.

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